higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

WORKSHOP PRACTICE
NQF LEVEL 2

20 NOVEMBER 2019

This marking guideline consists of 6 pages.

Copyright reserved

QUESTION 1

1.1
- Missing protection covers can cause workers get caught in the machinery
- Overcrowding can lead to stress and unhappiness
- Inadequate lighting increases human error
- Inadequate ventilation can cause illness and drowsiness

(Any TWO of the above or relevant answers) (2)

1.2
- Thermal requirements
- Lighting
- Windows
- Ventilation
- Housekeeping
- Noise (6)

1.3
1.3.1 often
1.3.2 three
1.3.3 time
1.3.4 place

(4 × 1) (4)

1.4
- Step 1: Establish the effort rating (score %)
- Step 2: Calculate the DIFR
- Step 3: Determine and allocate the safety, health and environmental frequency severity
- Step 4: Apply criteria and establish final star grading (4)

1.5 20 employees. (1)

1.6
- Long-sleeve overall
- Leather apron
- Leather gloves
- Leather spats
- Welding helmet/shield (Any THREE of the above or relevant answers) (3)

[20]

QUESTION 2

2.1 The casualty should:
- be able to breathe
- have enough oxygenated blood circulating through the body (2)

2.2 By applying:
- hard back slaps
- chest compressions (chest thrusts)
- the Heinlich Manoeuvre (this is not recommended for a casualty who is under ten years of age) (3)

2.3
- The S.A. Red Cross Society
- The St Johns Ambulance
- The S.A. First-Aid League
- A person or organisation approved by the Chief Inspector for this purpose (4)

2.4
- Through the digestive tract✓ – by substances being swallowed.✓
- Through the skin✓ – by substances being injected into the bloodstream.✓
- Through the respiratory tract✓ – by gases or fumes being inhaled.✓

(3 × 2) (6)

[15]

QUESTION 3

3.1 The point of a centre punch must be held at 90˚. (1)

3.2 If the pin punch is bent and the ends have burrs, the punch will malfunction. (2)

3.3 3.3.1 Used to mark off holes that must be drilled.

3.3.2 Used to drive out pins and rivets.

3.3.1 Used for cutting holes in gaskets bolts or studs to pass through.

(3 × 2) (6)

3.4. 3.4.1
- Ball peen/Engineering hammer
- Claw hammer
- Soft-face hammer
- Sledge hammer
- Chipping hammer (Any TWO of the above) (2)

3.4.2
- Hold the hammer firmly at the end of the handle/shaft
- For light hammering use wrist only
- Use the entire arm for heavy blows
- Use the bounce of the hammer for upward strokes (4)

[15]

QUESTION 4

4.1
- Portable hand-held drilling machine
- Angle grinder
- Orbital sander
- Electric screwdriver
- Jigsaw (Any FOUR of the above or relevant answers) (4)

4.2.1
- Store grinding wheels carefully on a wooden rack to prevent damage to it
- Place small wheels in suitable boxes to prevent them from bumping against one another
- The store room must be dry and protected from extreme temperatures or vibration (3)

4.2.2
- Periodically lubricate the gear-and-rack table elevation mechanism, the grooves in the spindle, and the rack
- Ensure that the base is properly secured to the workshop floor
- Keep the column at all times (3)

4.3.1 Is used to make cut-throughs and cut-outs in wood, plastic and metals (2)

4.3.2 Blade (1)

4.3.3 Safety goggles and a dusk mask (2)

[15]

QUESTION 5

5.1.1 Vernier calliper (1)

5.1.2
- The vernier calliper's jaws must move freely and firmly. The main scale and vernier sliding scale must be legible
- The locking screw and thumb wheel must be operational
- Make sure the enclosure is in good condition
- The vernier calliper must be kept dust free (Any FOUR of the above) (4)

5.2
- Have different lengths of measuring tapes available
- Do not unroll the entire measuring tape to measure just a few millimetres.
- If possible, ask an assistant to hold the tape on one side.
- Always reel in the measuring tape in steps, and keep your hands away from the tape. (Any THREE of the above or relevant answers) (3)

5.3
- 2 m
- 3 m
- 5 m
- 10 m (4)

5.4
- The steel tape must be stored in a rolled-up position
- The steel tape must be placed in its own sectional area in the toolbox
- The steel tape can be placed on a bracket on the tool board in the workshop (3)

[15]

QUESTION 6

6.1 The purpose of symbolic signs is to achieve instant recognition and understanding✓ by all language and ethnic groups,✓ regardless of their degree of literacy.✓ *(*Any relevant answer) (3)

6.2
- Careless people
- Electricity
- Spontaneous combustion
- Chemicals
- Unsafe handling of flammable liquids and gases (5)

6.3 The Weston pulley block. (1)

6.4
- Pulley and chain
- Manual chain block
- Electric chain block (Any ONE of the above) (1)

[10]

Copyright reserved

QUESTION 7

7.1
- The voltage rating
- The wattage rating
- The temperature control
- If it has anti-static protection
- The different types of bits available (soldering iron tips) (5)

7.2
- Immediately cool the affected area with cold running water for a few minutes
- Remove any rings before swelling starts
- Apply a sterile dressing to the wound to protect it against infection
- Do not apply lotions or ointments or prick any blisters that form later
- Seek professional medical advice when necessary (5)

[10]

TOTAL: 100

Copyright reserved